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IR TrueFISP With a Golden-Ratio-Based Radial Readout: Fast Quantification of T_1, T_2, and Proton Density

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A promising approach for the simultaneous quantification of relative proton density (M_0), T_1, and T_2 is the inversion-recovery TrueFISP sequence, consisting of an inversion pulse followed by a series of balanced steady-state free precession acquisitions. Parameters can then be obtained from a mono-exponential fit to the series of images. However, a segmented acquisition is usually necessary, which increases the total acquisition time considerably. The goal of this study is to obtain M_0, T_1, and T_2 maps using a single-shot acquisition, with T_1 and T_2 measurements in brain that are consistent with the published literature, with a 20-fold speed improvement over the segmented approach, and at a clinically relevant spatial resolution. To this end, a single-shot inversion-recovery TrueFISP sequence was combined with a radial view-sharing technique. The parameters M_0, T_1, and T_2 were then obtained on a pixel-wise basis from a three fit parameter to the signal evolution. The accuracy of this method for quantifying these parameters is demonstrated in vivo. In addition, further corrections to the quantification necessary owing to other experimental factors, namely magnetization transfer and imperfect slice profiles, were developed. Including additional scans necessary for these corrections in the measurement protocol, the required scan time is increased from approximately 6 to 18-28 s per slice. **Magn Reson Med 69:71–81, 2013. © 2012 Wiley Periodicals, Inc.**

Key words: IR TrueFISP; TrueFISP; bSSFP; synthetic image generation; T_1 mapping; T_2 mapping; inversion recovery

Quantitative magnetic resonance imaging (MRI) involves pixel-wise mapping of proton density, longitudinal relaxation time T_1, or transverse relaxation time T_2 (or other relevant parameters) at each location in the tissue to be characterized. There has been significant recent interest in a quantitative approach to MRI, as it provides a means of evaluating pathology using absolute tissue characteristics rather than a contrast-based approach (1–7). In addition, it has long been recognized that if these parameters can be mapped in a time-efficient manner, theoretically images of any desired contrast could be retrospectively generated, and several groups have contributed to the literature on this long-standing goal in MRI (8–12).

Many different quantification techniques have been proposed in the previous literature. In most cases, T_1 and T_2 values are determined using two separate experiments (proton density is often provided by both). The most common methods for T_1 mapping are either based on the inversion-recovery (IR) spin-echo or gradient-echo sequences (13,14), or rely on multiple scans with variable flip-angle (15,16) or repetition times (TR). For T_2 mapping, the gold-standard method is to exponentially fit the T_2 decay to the signal from multiple spin-echo experiments with variable echo times. Often, to reduce scan time, a multi spin-echo sequence is used instead (17,18). Recently, another very fast method for T_1 and T_2 determination that uses the steady-state signal level from multiple spoiled and balanced steady-state free precession scans with variable flip angles for T_1 and T_2 quantification has been proposed (19). However, it is not trivial to accelerate this method further using view-sharing techniques, as it does not rely on fine sampling of a transient signal along the temporal dimension, but on several image acquisitions in the steady state.

Another promising approach for the *simultaneous* quantification of proton density, T_1 and T_2, that captures a smooth transition after signal preparation, is the IR TrueFISP sequence (20). This sequence consists of an inversion pulse followed by the acquisition of several balanced steady-state free precession (bSSFP) images as the signal time course approaches the steady state. The parameters can then be obtained from a mono-exponential three fit parameter to the series of images. However, for accurate quantification it has been necessary to acquire this time series in a segmented fashion, owing to the rapid relaxation of the signal toward steady state. The need for such a segmented approach considerably increases the required acquisition time. If one were to use this segmented approach for generating synthetic images corresponding to a standard clinical exam, this approach would still be slightly slower than the conventional exam. To facilitate a clinical transition to a quantitative imaging approach, it is therefore desirable to generate the quantitative maps, and by extension all the images needed for a conventional exam, with an outlay of time that is equal to or less than that required presently for the conventional approach of separately acquired image sequences. Our goal was therefore to obtain proton density, T_1 and T_2 maps with a clinically acceptable spatial

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resolution and in an acceptable time frame. To this end, we combined the IR TrueFISP-based parameter mapping with a radial trajectory with golden-ratio-based profile (21). The advantage of this radial approach is that the center of *k*-space, and therefore the information about the image contrast, is frequently updated during scanning. Using a suitable view-sharing *k*-space filter, we show that it is possible to characterize the evolution of the signal toward the steady state with high-temporal resolution. As in a conventional IR TrueFISP, proton density, T_1 and T_2, can then be obtained from a fit to the image series.

The aim of this study is to develop an accurate and fast T_1 and T_2 relaxometry method that has the potential for clinical adoption. This manuscript describes the generation of proton density, T_1, and T_2 maps from IR TrueFISP data acquired along a golden angle radial trajectory. These data can be acquired in <6 s per imaging slice, meeting the requirements of short scan time discussed above. Quantification errors of the IR TrueFISP method owing to slice profile imperfections and magnetization transfer (MT) effects are examined. Corrections for these effects are presented to yield more accurate values.

THEORY

IR TrueFISP-Based Parameter Estimation

For on-resonant spins, the IR TrueFISP signal after *n* TRs can be described by a three-parameter exponential, according to (20):

$$S(n \cdot \mathrm{TR}) = S_{\mathrm{StSt}} - (S_0 + S_{\mathrm{StSt}}) \cdot \exp((-n \cdot \mathrm{TR})/T_1^*) \quad [1]$$

where S_{StSt} is the steady-state signal; S_0 is the transient state signal extrapolated to $t = 0$; and T_1* is the apparent relaxation time; T_1, T_2, and M_0 can then be obtained from the three fit parameters S_{StSt}, S_0, and T_1*, and the excitation flip angle α, with the assumption that TR $\ll$ T_2, according to (20,22):

$$T_1 = T_1^* \frac{S_0}{S_{\mathrm{StSt}}} \cos\,\alpha/2 \quad [2]$$

$$T_2 = T_1^* \sin^2\alpha/2 (1 - \frac{S_{\mathrm{StSt}}}{S_0} \cos\,\alpha/2)^{-1} \quad [3]$$

$$M_0 = \frac{S_0}{\sin\,\alpha/2} \quad [4]$$

Eqs. 2 and 3 illustrate that quantification of T_1 and T_2 is correlated in an IR TrueFISP experiment. Specifically, flip angle errors can lead to information leakage from T_1 to T_2 and vice-versa. Thus, correct knowledge of the (effective) flip angle is necessary for accurate quantification.

Golden-Ratio-Based Profile Order

In conventional radial MRI, *k*-space is sampled with equidistantly spaced radial lines in the order of increasing azimuthal angles from 0 to π (or 2π). For dynamic applications, this profile order is often interleaved so that view sharing can be employed (23). Recently, a radial profile order was proposed, with a constant azimuthal angle spacing of approx. 111.246°, or π divided by the golden ratio (21). This profile order has the advantage of a near-optimal uniform *k*-space sampling for any number of projections, whereas an interleaved acquisition is only optimal for a predetermined number (specifically the total number of projections divided by the number of interleaves). This flexibility makes it possible to choose the frame rate of a dynamic acquisition retrospectively and/or dependent on the position in *k*-space. Owing to the close relationship between the golden ratio and the Fibonacci series, *k*-space sampling becomes most uniform whenever the number of projections approaches a Fibonacci number.

Image Reconstruction Using a Modified *k*-Space-Weighted Image Contrast Filter

Using view sharing, the golden-ratio-based profile order makes it possible to generate a large number of images with different contrasts from a single-shot IR experiment. To reconstruct a single time frame, the center of *k*-space is taken from only a few acquired projections close to the chosen reconstruction time point, whereas the high spatial frequencies are provided by including data from time points further away, as shown in Fig. 1a. Such a *k*-space filter, as initially proposed by Song and Dougherty (24) by the name of *k*-space-weighted image contrast, ideally leads to a uniformly sampled *k*-space for every time frame, thereby avoiding undersampling artifacts. As shown in Fig. 1b, the reconstruction window starts with a given number of projections for the center of *k*-space (e.g., 8), and increases to the next Fibonacci number when the Nyquist sampling criterion would otherwise be violated. The next time frame is then reconstructed by moving this *k*-space filter by a chosen number of projections. In this study, the filter was moved in increments of eight projections, yielding a temporal resolution of between 32 and 64 ms, depending on the TR of the sequence.

For density compensation, a standard Ram-Lak DCF was used that was corrected for angular anisotropy owing to the golden-ratio-based profile order. To this end, the angular distance to the two next neighboring projections was determined for each projection in each ring of the radial *k*-space filter (Fig. 1b) and the Ram-Lak DCF was weighted by this distance. The image time series was then reconstructed using the *nonuniform fast Fourier transform* from the image reconstruction toolbox by Fessler and Sutton (25).

Slice Profile Effects

As the nominal flip angle is only reached in the center of the excited slice (assuming a perfectly homogeneous B_1 field), the actual flip angle in most parts of the slice is significantly smaller.

Using the nominal flip angle for 2D IR TrueFISP, quantification will therefore obviously lead to errors. We propose to correct for nonrectangular slice profiles by introducing an effective flip angle. The effective flip angle is defined as the flip angle that best describes the IR TrueFISP signal evolution for a given T_1 and T_2, assuming a perfect rectangular pulse profile ($\alpha_{\mathrm{effective}}(T_1,T_2) < \alpha_{\mathrm{nominal}}$). This parameter can be obtained from a simulation of the IR TrueFISP

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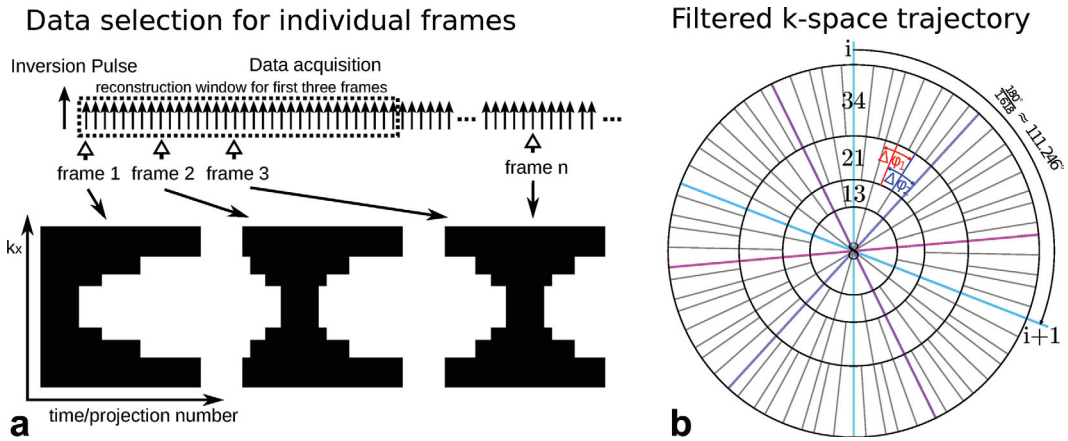


FIG. 1. **a**: Illustration of data selection by temporal *k*-space filtering. Initially, following the inversion pulse the filter is highly asymmetric, as there is no data from prior time points. Later, the filter becomes more and more symmetric. **b**: Illustration of golden-ratio-based radial profile order and filtering in *k*-space. Subsequent projections are spaced by an angle increment of 111.246° (five successive projections are color coded from cyan to magenta). For *k*-space filtering, projections are grouped into rings. The central ring consists of eight projections that were acquired closest to the target time point. To fulfill the Nyquist criterion, the included number of projections jumps to the next Fibonacci number further out in *k*-space, thus forming the next ring. This results in a relatively uniform *k*-space sampling density. This figure also shows that in each ring, the angular distance between the next neighbors of a projection alternates between two possible values (denoted by $\Delta\phi_1$ and $\Delta\phi_2$). Thus, a density compensation function based on the Ram-Lak filter has to be corrected for this angular anisotropy. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

experiment using a known (i.e., calculated or experimentally determined) slice profile, as described in the Materials and Methods section.

MT Effects

Another potential problem in mapping M_0, T_1, and T_2 with an IR TrueFISP experiment is that on-resonance MT effects can have a significant influence on the steady-state signal of bSSFP sequences (26), and can thus potentially confound the experiment.

The effects of MT on an IR TrueFISP experiment are twofold: First, it leads to a reduction of the apparent relaxation time (and fit parameter) T_1*. Second, it reduces the steady-state signal S_{StSt}. Quantification of proton density and of the signal magnitude directly after inversion, S_0, should not be significantly affected by MT because the saturation of the semi-solid spin pool at the beginning of the experiment is effectively 0. Thus, to correct T_1 and T_2 for MT effects, it is necessary to extrapolate S_{StSt} and T_1* to a MT-free state.

As previously proposed (27), reducing RF power by prolonging the RF pulse in a bSSFP acquisition can be effective in reducing MT-related signal loss. Stretching the RF pulse by a factor of β reduces the MT saturation rate per pulse by $1/\beta^2$, provided that the bandwidth of the pulse is much smaller than the bandwidth of the restricted spin pool causing MT (and assuming a constant flip angle) (27).

Gloor et al. (28) recently presented a modified bSSFP steady-state signal equation based on a two-pool MT model. However, full quantification of all relevant MT parameters using this model requires a large number of bSSFP scans in the steady state. The associated increase in scan time may be acceptable if the quantified MT parameters are of further interest, but would otherwise largely diminish the speed advantage of the IR TrueFISP method. Bieri and Scheffler (27) used a simple exponential model to describe the MT-induced steady-state signal change, and showed that the experimental data are well described using this model. Thus, we propose to correct the steady-state signal for MT using an exponential saturation model of the same general form:

$$\begin{aligned} S_{\mathrm{StSt}}(\beta) &= (S_{\mathrm{StSt}}^{\text{no MT}} - S_{\mathrm{StSt}}^{\text{full MT}}) \cdot (1 - e^{-k\cdot\beta}) + S_{\mathrm{StSt}}^{\text{full MT}} \\ &= \Delta S_{\mathrm{StSt}} \cdot (1 - e^{-k\cdot\beta}) + S_{\mathrm{StSt}}^{\text{full MT}} \end{aligned} \quad [5]$$

$S_{\mathrm{StSt}}^{\text{no MT}}$ denotes the steady-state signal extrapolated to $\beta \rightarrow \infty$ (free of MT effects), and $S_{\mathrm{StSt}}^{\text{full MT}}$ denotes the signal arising from a fully saturated semi-solid spin pool, i.e. the remaining observable signal depends primarily on the exchange rate between the spin pools and the T_1 relaxation time of the free spins. The rate constant k describes the exponential relaxation toward $S_{\mathrm{StSt}}^{\text{no MT}}$ with increasing β. By acquiring bSSFP images with at least three different pulse durations, it is possible to determine all three relevant parameters $S_{\mathrm{StSt}}^{\text{no MT}}$, $S_{\mathrm{StSt}}^{\text{full MT}}$, and k by performing a fit to Eq. 5. MT-related effects on the IR TrueFISP method can be reduced by using the extrapolated steady-state signal $S_{\mathrm{StSt}}^{\text{no MT}}$ in the calculation of T_1 and T_2.

The effect of MT on T_1* is generally smaller than on S_{StSt}. As each additional degree of freedom in the quantification leads to noise amplification, we choose to simply reduce MT-related effects by using T_1* from the experiment that was least affected by MT (i.e., from the scan with the highest β).

As S_0 should not be affected by MT, it is possible to use S_0 from any of the IR TrueFISP experiments for the final calculation of the relaxation parameters. We choose S_0 from the shortest TR scan, as it shows the least blurring owing to the higher sampling rate.

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METHODS

Imaging Experiments

Phantom experiments were performed on a 1.5-T whole-body imaging system (Magnetom Avanto, Siemens Healthcare, Erlangen, Germany) with a six-channel phased-array body matrix in combination with a spine matrix coil. Phantom experiments were performed to examine the validity of the radial IR TrueFISP and the slice profile correction method. To this end, seven 150-mL plastic bottles were filled with tap water and different amounts of a superparamagnetic iron oxide-based contrast agent (Resovist, Bayer Schering Pharma AG, Berlin, Germany). The concentration of the contrast agent was varied between 0 and $256 \cdot \times 10^{-6}$ M to yield T_1 values ranging from 316 to 2440 ms, and T_2 values ranging from 15 to 1680 ms. To reduce susceptibility-related artifacts, the phantom bottles were placed in a water-filled basin. A radial IR TrueFISP experiment with a golden-ratio-based profile order was performed with the following parameters: TR = 3.12 ms, flip angle = 45°, RF pulse duration (T_{RF}) = 0.8 ms, TBW = 2, matrix size = 128 × 128, field-of-view = 200 × 200 mm^2, slice thickness = 10 mm, receiver bandwidth = 1000 Hz/pixel, total number of projections acquired = 1800, total experiment time = 5.7s.

As a reference, the longitudinal and transversal relaxation times of the phantoms were measured with a Cartesian spin-echo sequence. Longitudinal relaxation was assessed from images acquired at five different inversion times (25/100/400/1200/5000 ms, TR = inversion time + 10 s, echo time = 15 ms, 4/5 rectangular FOV = 200 × 159 mm^2, matrix size = 128 × 77 [75% phase resolution]). After adiabatic inversion and subsequent inversion delay inversion time, a single *k*-space line was acquired, followed by a relaxation delay of 10 s before the next TR. Transverse relaxation was assessed from spin-echo images acquired with five different echo times (echo time = 13/50/150/400/800 ms, TR = 5 s, 4/5 rectangular FOV = 200 × 159 mm^2, matrix size = 128 × 77 [75% phase resolution]). Reference values for T_1 and T_2 were calculated from the corresponding phase-corrected real-part image series by three- and two-parameter fitting, respectively. For further analysis, a region-of-interest was drawn in each phantom bottle. Mean and standard deviation of T_1 and T_2 were then determined from these ROIs for the reference and the IR TrueFISP experiment, with and without correction for slice profile imperfections, which are described below.

To correct for slice profile imperfections, quantitative slice profiles for a sinc pulse (TBW = 2, FA = 45°, 60°) were obtained using a phantom containing a solution of 1.25 g $NiSO_4 \times 6H2O$ and 0.5 g NaCl per 1000 g distilled water (T_1 = 300 ms). The signal time course for the IR TrueFISP experiment was simulated for different T_1 and T_2 combinations and assuming on-resonant spins using Eq. 1 (nominal flip angle = 45°, 60°; TR = 5 ms) and depending on the position in the experimentally determined slice profiles. To find the effective flip angle, an estimate of the signal time course for a perfectly rectangular excitation profile is needed. This was taken to be the mean/sum across the excited volume. Using this assumption, the effective flip angle for a given T_1, T_2, and slice profile can then be obtained by fitting the sum of the signal over the slice to Eq. 1, with the effective flip angle as the only fitting parameter. A lookup table of effective flip angles was then generated for a wide range of T_1 and T_2 values (T_1 = 20–8000 ms; T_2 = 5–2000 ms). To reduce computation time, five times the expected apparent relaxation time T_1* was simulated to generate the lookup table. This table was used to correct the phantom and in vivo data for imperfect slice profiles, as described below.

In vivo experiments were performed on a 1.5-T whole-body imaging system (Magnetom Espree, Siemens Healthcare, Erlangen, Germany) with a 32-channel phased-array head receiver coil (Siemens Healthcare). IR TrueFISP experiments were performed on six asymptomatic volunteers. Informed consent was obtained and the project was approved by the local Institutional Review Board. After nonselective adiabatic inversion, a linearly increasing flip angle preparation consisting of four pulses was used to suppress off-resonance oscillations (29). After this preparation, the return to equilibrium was observed using a radial bSSFP sequence with a golden-ratio-based profile order.

To examine the influence of MT on parameter quantification, the IR TrueFISP experiment was repeated with different pulse durations and TRs. To keep the recently reported effect of finite RF excitation on the bSSFP signal (30) constant for all scans, the ratio between pulse duration and TR was kept fixed at 20%. The acquisition parameters used are as follows: TR = 4, 6, and 8 ms, pulse duration T_{RF} = 0.8, 1.2, and 1.6 ms (rf-scaling factor β = 1, 1.5, and 2), TBW = 2, flip angle = 60°, matrix size: 256 × 256, field-of-view = 220 × 220 mm^2, in-plane resolution = 0.86 × 0.86 mm^2, slice thickness = 6 mm, receiver bandwidth = 890 Hz/pixel. The total number of projections was adjusted so that the total acquisition time was just under 6 s for each experiment.

Signal Processing and Parameter Fitting

After the data were acquired, they were used to determine the first estimates of T_1, T_2, and M_0. First, for each individual time frame, adjacent projections were selected and filtered using the modified *k*-space-weighted image contrast scheme described above. Each frame contained eight projections in the center of *k*-space and reconstruction windows for consecutive frames were separated by these eight projections. Moving further out in *k*-space, the number of projections jumps to the next Fibonacci number (i.e., 13, 21, 34, 55, etc., up to a maximum of 233 projections), whenever the Nyquist criterion would be violated otherwise. Images were reconstructed from these filtered projections using nonuniform fast Fourier transform gridding (25) and an analytically calculated density compensation function as described in the Theory section. Multichannel images were combined using the adaptive combine method of Walsh et al. (31). The resultant time series of images was used to determine the values for the parameters M_0, T_1, and T_2 by fitting the data pixel-wise to Eq. 1.

After the determination of the values of M_0, T_1, and T_2 for each of the different pulses described above, the next

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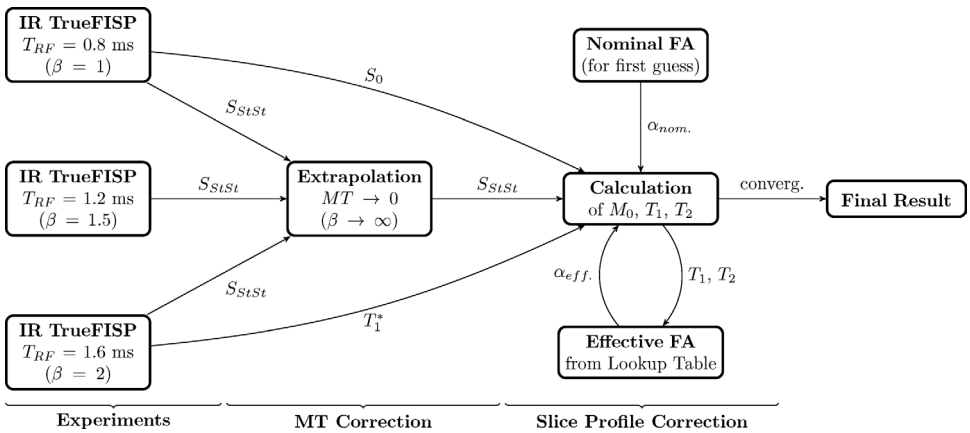


FIG. 2. Flow chart of the proposed corrections for IR TrueFISP quantification: After running three separate IR TrueFISP experiments with varying rf pulse durations and after performing fits to Eq. 1, the steady-state signal is extrapolated to the MT-free state. Using this extrapolation together with T_1* from the scan that was least affected by MT, and S_0 from one of the scans (using S_0 from the scan with shortest TR reduces blurring), first estimates of T_1 and T_2 are obtained that are iteratively corrected for slice profile imperfections using a lookup table in the following. Convergence is achieved when the mean change of the effective flip angle is below 1% between iterations.

step is to correct for MT effects and slice profile imperfections. A flow chart showing how these corrections were performed is given in Fig. 2. Using the steady-state signals from the three different pulse lengths, the MT-free signal $S_{\mathrm{StSt}}^{\mathrm{no\,MT}}$ was determined using the exponential model (Eq. 6). MT-corrected T_1 and T_2 maps were then calculated as described in the Theory section, using the corrected steady-state signal $S_{\mathrm{StSt}}^{\mathrm{no\,MT}}$, T_1* from the scan that was least affected by MT (β = 2), and the proton density map from the scan with the shortest TR (TR = 4 ms). The result of this correction is a set of M_0, T_1, and T_2 maps that are less affected by MT.

After these initial estimates of the MT-free T_1 and T_2 values are determined using the nominal flip angle, parameter maps were iteratively corrected to account for the imperfect slice profile with a two-step procedure. In the first step, the nominal flip angle is assumed to be valid over the whole slice and initial estimates of T_1 and T_2 values are calculated. In the second step, an estimate of the effective flip angle is obtained from the lookup table using the initial estimates of T_1 and T_2. Proton densities, T_1 and T_2, were then corrected and the effective flip angle was updated using the corrected T_1 and T_2 values. This procedure was iterated four times until convergence was achieved (mean effective flip angle changes by <1%). Total computation time for the iterative slice profile correction was under 0.1 s per slice on a standard desktop computer running unoptimized Matlab code.

RESULTS

To correct for slice profile imperfections, the signal evolution for an IR TrueFISP experiment was simulated using the experimentally determined slice profile arising from a sinc pulse ($\alpha_{\mathrm{nominal}}$ = 60°, TBW = 2) for spins with the following relaxation times: T_1 = 600 ms, T_2 = 60 ms. The signal time course as a function of the position in the slice is shown in Fig. 3a. Figure 3b shows the integral of this signal over the slice, as well as an exponential fit used to determine the effective flip angle. The effective flip angle for a nominal flip angle of 60° and the range of relaxation times described above, derived from performing the simulation with different relaxation parameters, is shown in Fig. 3c. Figure 3d shows the iteratively determined map of effective flip angles for one of the six volunteers. As can be seen, the variations of the effective flip angle between different brain tissue types are relatively small.

Representative images from an IR TrueFISP time course are shown in Fig. 4a. These images (taken from 187 total frames) were reconstructed using the *k*-space-weighted image contrast filter described above, and were used to obtain the parameter maps shown in Fig. 4b. Figure 4b shows the resulting parameter maps when no slice profile or MT corrections have yet been applied.

Figure 5 shows an overview of the results from the IR TrueFISP phantom experiments, summarized in more detail in Table 1. As shown in Fig. 5, the reference T_1 and T_2 values are plotted against the T_1 and T_2 values derived from the IR TrueFISP experiment. Results with and without slice profile correction are shown to demonstrate the effects of including this correction method. No MT correction was required for these phantom data, and thus it was not applied.

The effect of MT on parameter quantification is shown in Fig. 6. The top row shows steady-state images for all three IR TrueFISP experiments, as well as a theoretically MT-free steady-state image, obtained by extrapolation. The two bottom rows show the resultant T_1 and T_2 maps, which illustrates that MT leads to an overestimation of T_1 and underestimation of T_2. In white matter, T_1 is overestimated by up to 30%, and T_2 underestimated by up to 50% (depending on RF power).

Figure 7 shows the result of the parameter mapping with corrections for the slice profile and MT effects for two of the normal volunteers. To provide a quantitative comparison, white matter, gray matter, and cerebral spinal fluid (CSF) regions of interest in each volunteer were selected (shown in red in the proton density map of Fig. 7; similar regions of interest were selected for maps

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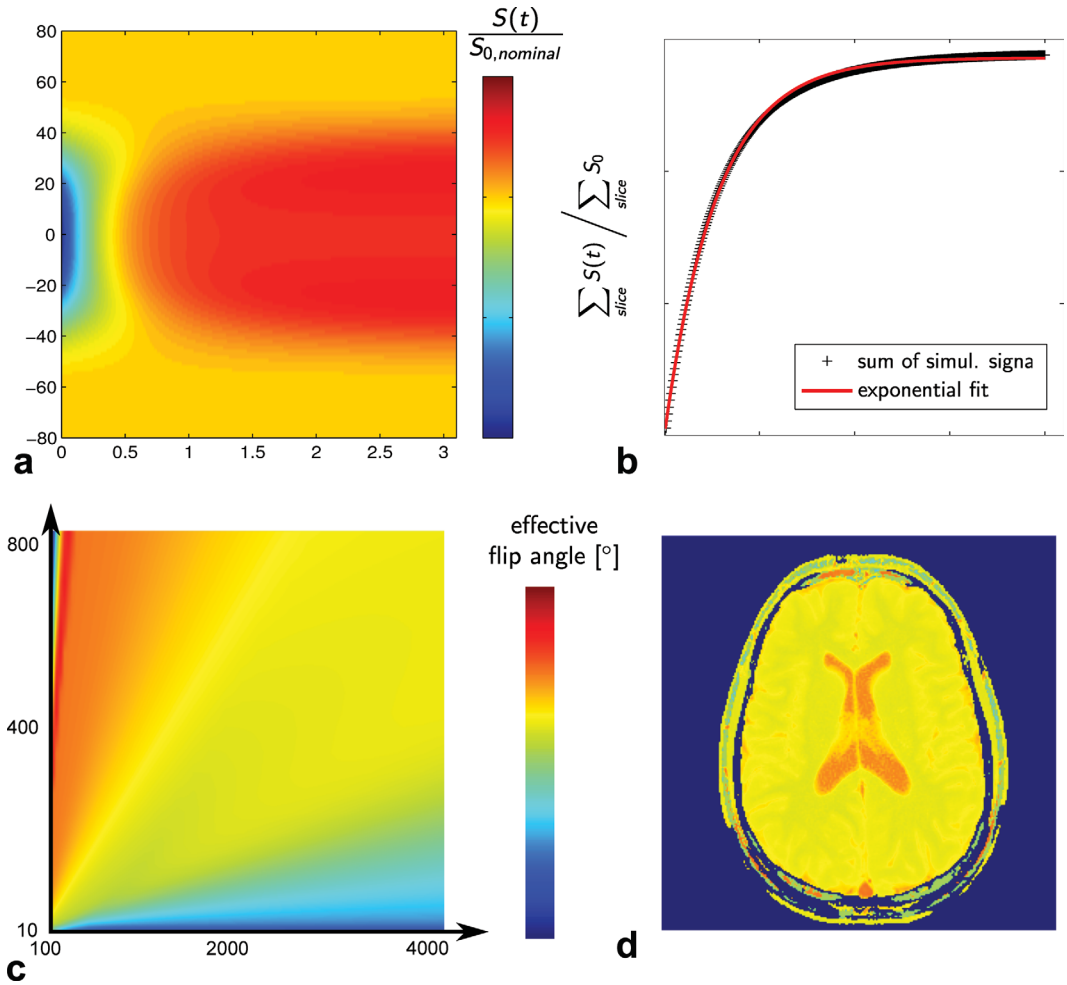


FIG. 3. **a**: Simulated signal evolution for an IR TrueFISP experiment depending on slice position (T_1/T_2 = 600/60 ms, sinc pulse with TBW = 2; slice thickness = 50 mm; nominal flip angle = 60°), (**b**) sum over slice of simulated signal and exponential fit through simulated data points. **c**: Effective flip angle lookup table determined from the fit for different T_1, T_2 combinations (d) Iteratively determined map of effective flip angles for one of the volunteers (same scaling as c). [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

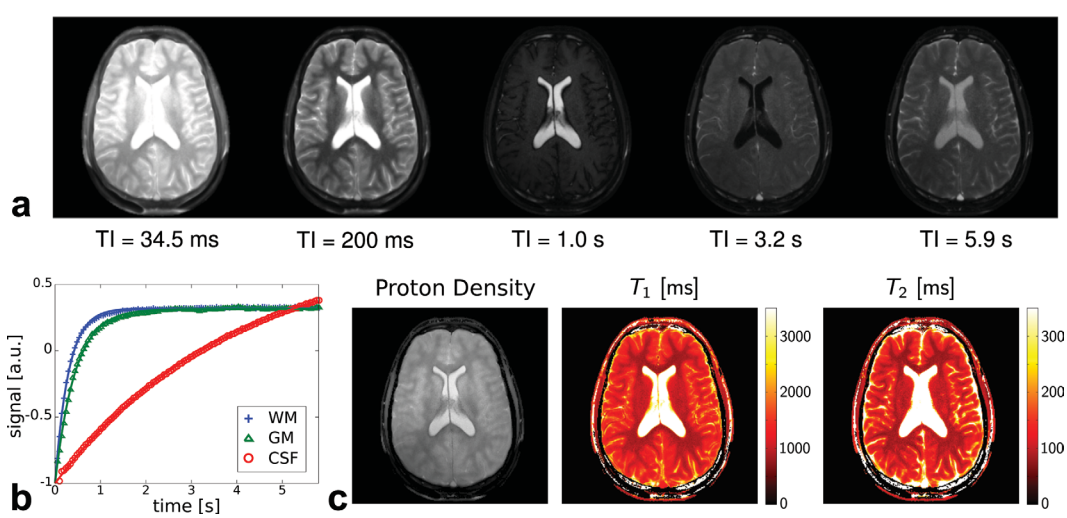


FIG. 4. **a**: Five representative time frames with inversion times ranging from 34.5 ms to 5.9 s from one of the IR TrueFISP experiments (TR = 4 ms). **b**: Reconstructed signal time course and fit for regions of interest in white matter, gray matter, and CSF. **c**: Proton density, T_1 and T_2 maps, determined from a fit to the reconstructed time series. These maps were corrected neither for slice profile nor for MT effects. The lower signal in the lateral part of the brain, visible in the proton density map, is caused by an incomplete compensation of sensitivity variations of the 32-channel receiver coil. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

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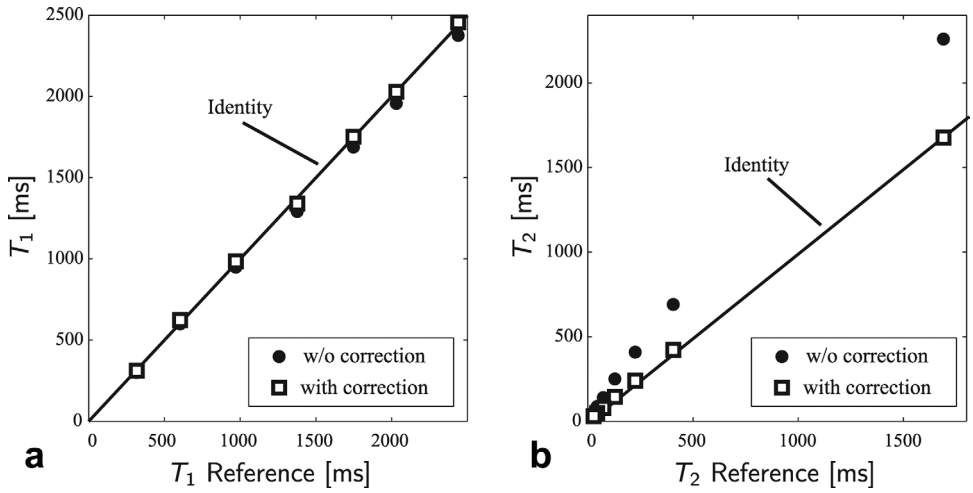


FIG. 5. Phantom results: T_1 and T_2 values from all phantoms with and without correction for imperfect slice excitation over the corresponding reference values (for a better distinction of short T_2 species, see Table 1). Magnetization transfer plays no role in the phantom experiments, so there was no need for an MT correction.

generated from the other volunteers). The T_1 and T_2 values, corrected for slice profile and MT, obtained from each region are listed in Table 2. In one case, it was not possible to find a region-of-interest containing pure CSF owing to through-plane partial volume effects, leading to the exclusion of this data set in the calculation of the mean CSF relaxometry data.

DISCUSSION

Although parameter mapping with an IR TrueFISP-based scheme has been previously demonstrated (20,22), the introduction of a golden-ratio-based radial readout with a temporal *k*-space filter greatly accelerates this technique, bringing the time required for parameter estimation down to 6 s/slice. This represents a 20-fold improvement over the previously reported, 2 min/slice obtained with a segmented Cartesian acquisition (20,22). This critical improvement in the speed of the parameter mapping experiment allows whole head coverage (20–30 slices) in 2–3 min, making quantitative and synthetic imaging clinically viable. With this acceleration, imaging time for IR TrueFISP parameter mapping is very similar to the combined DESPOT1/DESPOT2 approach which requires approximately 7.9 s per slice in an early implementation [19] (more precisely, parameter mapping for a 256 × 256 × 128 3D volume was achieved in 16 min and 48 s). A comparison of strengths and weaknesses of these two options for fast T_1 and T_2 mapping is left for a future study.

The role of parallel imaging in IR-TrueFISP-based parameter mapping is an important future consideration. CG-SENSE (32), for example, could be employed instead of *k*-space-weighted image contrast view sharing. Combining view sharing with a parallel imaging technique such as CG SENSE (32) or through-time Radial GRAPPA (33) is promising because it would potentially offer further acceleration beyond view sharing. Early study on this possibility is promising (34) but has not been published in manuscript form. Further study is needed to fully exploit parallel imaging acceleration.

However, deviations from traditional theory owing to MT effects and imperfect slice profiles lead to considerable quantification errors in the IR TrueFISP experiment, independent of the chosen *k*-space sampling scheme. Thus, corrections for these two effects must be included in the IR TrueFISP quantification technique, which increases the scan time from 6 to 28 s/slice including a relaxation delay of 5 s between IR TrueFISP experiments. The extra time owing to the relaxation delay can be avoided by using an interleaved multislice acquisition, reducing the scan time to 18 s/slice.

The influence of MT on parameter quantification is especially apparent when using high-power pulses (high flip angle and high bandwidth) and short TRs. In an IR TrueFISP experiment, the MT builds up simultaneously with the relaxation of the signal toward the steady state, potentially distorting parameter quantification. We examined the effect of MT on parameter quantification by varying the duration of the RF pulse, as proposed

Table 1
Phantom Data: T_1 and T_2 for all Seven Phantom Bottles, With and Without Correction for Imperfect Slice Excitation, and Corresponding Reference Values (Mean ± SD)

	T_1 (ms)						
	Bottle 1	*Bottle 2*	*Bottle 3*	*Bottle 4*	*Bottle 5*	*Bottle 6*	*Bottle 7*
W/o corr.	2376 ± 12	1958 ± 11	1689 ± 11	1292 ± 13	949 ± 14	600 ± 14	300.0 ± 9.3
With corr.	2456 ± 12	2029 ± 11	1752 ± 11	1341 ± 13	985 ± 14	623 ± 14	312 ± 10
Reference	2439 ± 16	2030 ± 8	1747 ± 8	1376 ± 8	971 ± 9	604 ± 12	317 ± 15
	T_2 (ms)						
W/o corr.	2260 ± 250	691 ± 29	409 ± 13	251 ± 10	140.6 ± 5.3	87.1 ± 4.8	53.4 ± 6.5
With corr.	1680 ± 240	420 ± 20	240.1 ± 8.1	144.3 ± 6.1	79.2 ± 3.1	48.7 ± 3.8	30.4 ± 4.1
Reference	1678 ± 59	394.2 ± 7.7	213.4 ± 4.8	116.6 ± 2.4	61.7 ± 1.6	32.2 ± 1.2	15.4 ± 1.9

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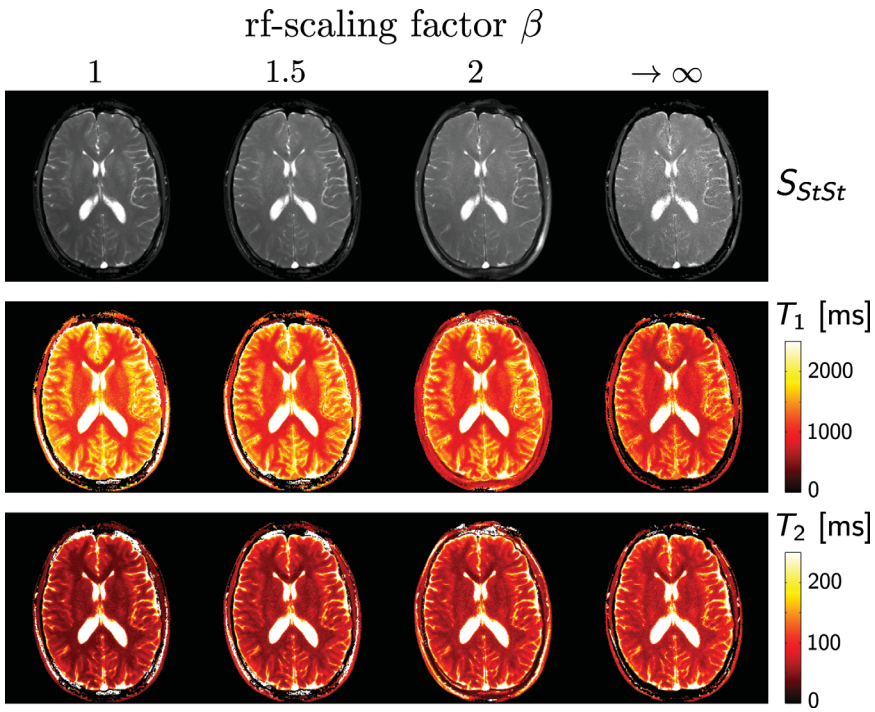


FIG. 6. From top to bottom: steady-state signal, T_1, and T_2 maps, as obtained from a three-parameter fit, for three experiments with different rf-scaling factors β and thus MT levels (first three columns), as well as extrapolated to $\beta \rightarrow \infty$ (far right column). Short pulses (small β) lead to a severe attenuation of the steady-state signal owing to an increased MT effect, which translates into an overestimation of T_1, and underestimation of T_2. Quantification of M_0 is not affected by MT (and is thus not shown in this figure). [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

previously (27). However, to limit the recently reported influence of finite RF excitation on bSSFP (30), it was necessary to keep the T_{RF}/TR ratio fixed at a relatively low value, which limited the feasible range of T_{RF} values and thus the dynamic range that was available for MT extrapolation. Nonetheless, extrapolation of the steady-state signal toward the MT-free state was relatively stable although it did lead to a noticeable noise increase in the MT-corrected parameter maps (compare raw parameter maps with corrected [extrapolated] maps on the far right of Fig. 6). The influence of finite RF excitation on parameter quantification, and thus the necessity to keep the T_{RF} /TR ratio low, may be avoided in future study using the recently proposed concept of superbalanced steady-state free precession (35). This concept allows the use of much longer RF pulses by overcoming the problem of finite RF effects, potentially eliminating the need for MT correction based on an extrapolation from multiple experiments.

Owing to the relatively low influence of MT-related T_1* deviations on parameter quantification and to avoid further SNR loss as a result of additional degrees of freedom, T_1* was not extrapolated to a MT-free state. MT effects on T_1* were simply reduced by using the value from the scan that was least affected by MT (highest β). Extrapolation of T_1* from a region of interest in white matter, the"worst case" scenario for MT in the brain, toward the MT-free state suggests that this leads to an underestimation of T_1* of approximately 3%, leading to similar underestimations of T_1 and T_2. In principal, this approach requires only two traditional bSSFP acquisitions in the steady state with different RF pulse durations, in addition to the fit parameters from a single IR TrueFISP experiment. This strategy is more time-efficient than performing three separate IR TrueFISP experiments, reducing the acquisition time penalty of the proposed MT correction method: Assuming that one steady-state

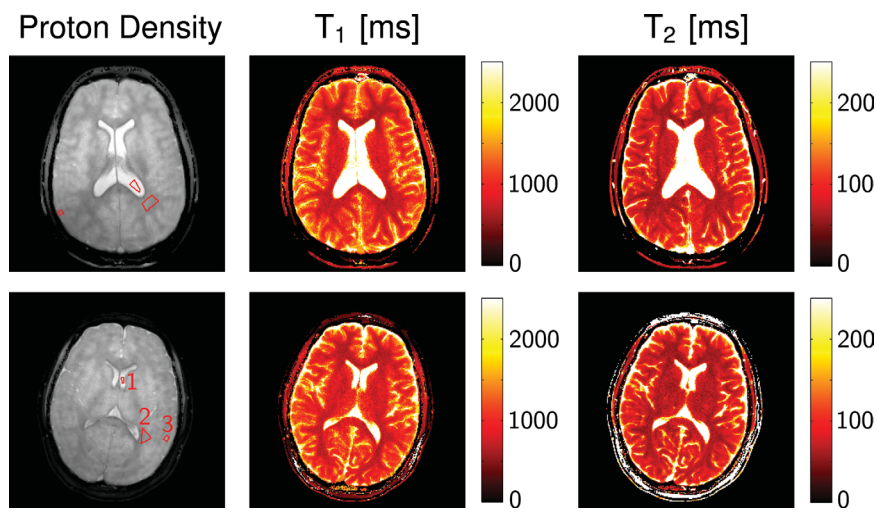


FIG. 7. Parameter maps, corrected for slice profile and MT, for two of the normal volunteers. Regions of interest used for further analysis (Table 2) are shown in the proton density maps in red; 1: CSF; 2: white matter; 3: gray matter. [Color figure can be viewed in the online issue, which is available at wileyonlinelibrary.com.]

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Table 2
Volunteer Data: Parameters Obtained from ROIs Positioned in Different Parts of the Brain (Mean ± SD over Six Volunteers/Five Volunteers in Case of CSF), Corrected for Slice Profile Inaccuracies and Magnetization Transfer

	T_1 (ms)	T_2 (ms)
White matter (dorsal)	716 ± 23	65.0 ± 3.5
Previously reported	*608–756*	*54–81*
Gray matter	1123 ± 36	88.0 ± 4.5
Previously reported	*998–1304*	*78–98*
CSF	4200 ± 400	2100 ± 360
Previously reported	*4103 ± 904*	
	5127 ± 350	

image can be acquired in 3 s, this decreases the required imaging time per slice from 18 to 12 s. In many cases, it may also be viable to simply limit MT-related quantification errors by running a single IR TrueFISP experiment with relatively long RF pulses and low flip angles which results again in the originally stated imaging time of 6 s.

MT and slice profile-corrected quantitative results for ROIs in white matter, gray matter, and CSF are summarized in Table 2. Table 2 shows that the T_1 values in white and gray matters obtained from the radial IR TrueFISP method were 716 ± 23 and 1123 ± 36 ms (mean ± standard deviation between data from six volunteers), respectively, which are in good agreement with the published literature (6,19,36–40). Similarly, measured T_2 values were 65.0 ± 3.5 and 88.0 ± 4.5 ms in white and gray matters, respectively, again consistent with those reported previously in the literature (6,19,40,41). The obtained T_1 and T_2 values in CSF were 4200 ± 400 and 2100 ± 360 ms, respectively.

Figure 5 and Table 1 show that the values of the parameters obtained using this method in a phantom study are generally consistent with the values derived from spin-echo sequences. However, in vials with very high contrast agent concentrations and very low T_1 and T_2 values, T_2 is increasingly overestimated, even after correcting for slice profile imperfections (Table 1). This may be explained by the fact that the faster dynamics owing to the shortened T_1* (induced by short T_1 and T_2) cannot be fully captured by the temporal *k*-space filter. This is a well-known effect resulting from view sharing. If the signal changes too rapidly to effectively share information across frames, this will result in errors in the quantification. The very fast dynamics directly after the inversion pulse are temporally smoothed, leading to an underestimation of S_0. This then causes an underestimation of M_0 and overestimation of T_2 during the calculation of the relaxation parameters. In contrast, the influence of an underestimated proton density on the quantification of T_1 is rather small. Another potential contributing factor to the overestimation of T_2 in short T_2 species is the deviation of the bSSFP signal from traditional theory, derived using the assumption of instantaneous RF excitation (30,42). It is important to note that MT does not affect the phantom experiments as there is no restricted spin pool (the phantom did not contain a significant quantity of macromolecules).

The proposed correction for slice profile imperfections relies on the assumption that the sum of exponentials from different parts of the slice can be described by a single exponential. Simulation results show that this assumption holds true in good approximation for typical excitation profiles and physiological T_1/T_2 combinations (an example is shown in Fig. 3b). The validity of the slice profile correction is further supported by the results from the phantom experiments. As shown in Fig. 5a and Table 1, T_1 values obtained from the IR TrueFISP experiment are in excellent agreement with the reference at all seven contrast agent concentrations. After correcting for slice profile imperfections, the agreement is further improved. The necessity for slice profile correction is particularly obvious when looking at T_2 values in Fig. 5b. Without correction, T_2 values are substantially overestimated compared to the reference values. The proposed slice profile correction reduces this deviation from the reference, particularly for long T_2 species. It was observed that the effective flip angle shows only a small T_1/T_2 dependence in tissue. In this study, the effective flip angle was found to be between 42 and 44° for white and gray matters, and between 44 and 45° in CSF (for a nominal FA = 60°, TBW = 2, see Fig. 3d). As the effective flip angle is essentially constant regardless of tissue type, it could be reasonable to use a predetermined single effective flip angle for slice profile correction, instead of the iteratively determined map of effective flip angles. An alternative to slice profile correction is to move to a true 3D acquisition. The flip angle that is used in parameter calculation can then be adjusted for every voxel according to its position inside the excited slice (taking advantage of the additional spatial encoding in slice direction), making slice profile correction unnecessary.

An inhomogeneous B_1 excitation field will result in quantification errors that are similar to those discussed in the case of imperfect slice profiles (i.e., the errors will mainly be notable in the quantification of T_2 values). Fortunately, the B_1 excitation field is expected to be relatively homogeneous over the whole brain for the body coil (a typical birdcage coil) at field strengths of 1.5 T (∼ 64 MHz) and below (43). Thus, observation of quantification errors owing to B_1 field inhomogeneity was not a focus of this study. However, at higher field strengths, an additional B_1 field mapping scan may be necessary to achieve accurate T_2 quantification.

It is widely known that bSSFP is very susceptible to off-resonance frequencies. In addition to the obvious problem of banding artifacts, off-resonances can lead to errors in parameter quantification, by altering the steady-state signal and the transient signal response of the bSSFP sequence (44). This can be problematic for slices that are placed in the lower part of the brain, close to the nasal cavities and the ear canal. In this case, additional field mapping scans may be required to account for off-resonance effects. Off-resonance correction can potentially be achieved using a method similar to the proposed slice profile correction, i.e. iterative correction using a lookup table that was obtained from a simulation of the experiment. In the experiments shown in this study, the shim was relatively homogeneous throughout the slice, with an off-resonance frequency range of approximately ±20 Hz in brain tissue. Investigation of off-resonance effects on IR TrueFISP quantification is

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beyond the scope of this study and is thus left for a future study.

CONCLUSION

This study has shown that it is possible to derive T_1, T_2, and relative proton density from a single radial IR TrueFISP experiment in under 6 s per slice. This was achieved by using a golden-ratio-based profile order in combination with view sharing. Because the mathematical description of the ideal IR TrueFISP experiment used here does not take into account effects such as slice profile deviations, MT, off-resonance effects, B_1 inhomogeneities, and finite RF excitation, several corrections have been introduced. These corrections increase the total scan time per slice from 6 to 12–18 s. MT leads to a considerable overestimation of T_1 and underestimation of T_2 in these experiments. These errors can be effectively reduced by lengthening the RF pulse or partially compensated using the corrections shown here. Although this correction requires multiple scans and can lead to a loss in SNR owing to the additional degrees of freedom, the resulting scan time and image quality are still acceptable. Additionally, slice profile imperfections can lead to a significant overestimation of T_2. This effect was corrected by introducing an effective flip angle, derived from simulations of the experiment. The IR-TrueFISP technique has been tested in a phantom and in in vivo experiments and shown to yield parameters which are similar to those derived using standard quantification experiments. This study shows that the golden angle radial IR TrueFISP method is a viable option for quantitative MRI owing to significantly shorter scan times while still allowing relatively accurate parameter quantification.

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